

Invariant Gaussian Weighted Shifts on c_0 Have Unimodular Eigenvalues

Packet for arXiv:1112.1289

2026-06-20

Source Question

The source is Frederic Bayart and Etienne Matheron, *Mixing operators and small subsets of the circle*, arXiv:1112.1289. In Section 7.8, Question (7), the authors ask:

Is there an ergodic weighted shift on $c_0(\mathbb{N})$ with no unimodular eigenvalues?

Haar-null in X_p for any $p \in [1, \infty]$ if $|\lambda| > 1$.

7.8. Some questions. To conclude the paper, we list a few “natural” questions, most of which have already been raised.

- (1) Is Theorem 2.6 true without assuming that the family $\mathbf{S}$ is c_0 -like?
- (2) Is Theorem 2.8 true for $L^1(m)$ -mixing, or in the mild mixing case?
- (3) Let T be a hypercyclic operator on X whose $\mathbb{T}$ -eigenvectors span a dense subspace of X . Is T ergodic in the Gaussian sense? Is T frequently hypercyclic?
- (4) Let $T \in \mathcal{L}(X)$, and assume that for any set $D \subset \mathbb{T}$ with empty interior, the linear span of $\bigcup_{\lambda \in \mathbb{T} \setminus D} \ker(T - \lambda)$ is dense in X . Is it possible to find a countable family of continuous $\mathbb{T}$ -eigenvector fields $(E_i)_{i \in I}$, where each E_i is defined on some nontrivial closed arc $\Lambda_i \subset \mathbb{T}$, such that $\text{span}(\bigcup_{i \in I} E_i(\Lambda_i))$ is dense in X ?
- (5) Let H be a Hilbert space, and let $T \in \mathcal{L}(H)$ be mixing with respect to some Borel measure on H with full support. Is T mixing in the Gaussian sense?
- (6) On which Banach spaces is it possible to find ergodic operators (in the Gaussian sense) with no eigenvalues? The space X should of course not have cotype 2, but this is not enough: for example, if X is hereditarily indecomposable then there are no frequently hypercyclic operators at all on it, and hence no ergodic operators either ([48]). In fact, the question splits into two separate problems: (i) on which Banach spaces is it possible to find ergodic operators? (ii) what can be said if the space does not have cotype 2? We refer to [20] for general results regarding the first problem.
- (7) Is there an ergodic weighted shift on $c_0(\mathbb{N})$ with no unimodular eigenvalues?
- (8) Let $T \in \mathcal{L}(X)$ be ergodic in the Gaussian sense. Are the ergodic measures with full support dense in the set of all T -invariant measures (endowed with the usual Prokhorov topology)? See [49] for a positive answer in a completely different situation, and [18] for more in that direction.
- (9) Is it true that if T is an ergodic operator, then $HC(T)$ is Haar-null? Is this true at least for weighted shifts?

Answer

No. In fact, ergodicity is not needed: a weighted backward shift on $c_0(\mathbb{N})$ with an invariant Gaussian probability measure of full support must have unimodular eigenvalues.

Definitions and Conventions

Let $\mathbf{w} = (w_n)_{n \geq 1}$ be a bounded scalar weight sequence and let

$$B_{\mathbf{w}}(x_0, x_1, x_2, \dots) = (w_1 x_1, w_2 x_2, \dots)$$

act on $c_0(\mathbb{N})$. Put $W_0 = 1$ and $W_n = w_1 \cdots w_n$ for $n \geq 1$. The standard weighted-shift eigenvalue computation gives

$$B_{\mathbf{w}}x = \lambda x \iff x_n = \frac{\lambda^n x_0}{W_n} \quad (n \geq 0),$$

provided no weight before n is zero. Thus, for $|\lambda| = 1$, a nonzero eigenvector in c_0 exists exactly when

$$\frac{1}{W_n} \longrightarrow 0.$$

If some $w_n = 0$, no full-support invariant Gaussian measure can exist, since one coordinate of $B_{\mathbf{w}}x$ is then identically zero.

Proof Intuition

An invariant Gaussian measure carries a covariance matrix. The weighted shift does not leave the coordinate variances free: invariance forces each variance to be the previous variance divided by $|w_n|^2$. Since the measure lives on c_0 , the centered coordinates must converge to zero in probability, and a scalar Gaussian variable converges to zero in probability only when its variance goes to zero. Hence the products $|W_n|$ must tend to infinity, which is precisely the c_0 unimodular-eigenvector condition.

Lemma 1. *Let (Y_n) be scalar centered Gaussian random variables. If $Y_n \rightarrow 0$ in probability, then $\mathbb{E}|Y_n|^2 \rightarrow 0$.*

Proof. Write $s_n^2 = \mathbb{E}|Y_n|^2$. If $s_n \not\rightarrow 0$, then for some $\varepsilon > 0$ there is a subsequence with $s_n \geq \varepsilon$. Along that subsequence, write $Y_n = s_n g_n$, where each g_n has the fixed standard real or complex centered Gaussian law normalized by $\mathbb{E}|g_n|^2 = 1$. Then

$$\mathbb{P}(|Y_n| > \varepsilon/2) = \mathbb{P}(|g_n| > \varepsilon/(2s_n)) \geq \mathbb{P}(|g_n| > 1/2) > 0.$$

This contradicts convergence in probability to 0. □

Theorem 1. *Let $B_{\mathbf{w}}$ be a weighted backward shift on $c_0(\mathbb{N})$. If $B_{\mathbf{w}}$ admits an invariant Gaussian probability measure with full support, then $|W_n| \rightarrow \infty$. Consequently $B_{\mathbf{w}}$ has unimodular eigenvalues. In particular, there is no ergodic weighted shift on $c_0(\mathbb{N})$ with no unimodular eigenvalues.*

Proof. Let μ be an invariant Gaussian probability measure on $c_0(\mathbb{N})$ with full support, and let $X = (X_n)_{n \geq 0}$ be a c_0 -valued random variable with law μ . Let $m = \mathbb{E}X \in c_0$, and set

$$Y_n = X_n - m_n, \quad a_n = \mathbb{E}|Y_n|^2.$$

Each Y_n is a centered scalar Gaussian random variable.

Since $B_{\mathbf{w}}X$ and X have the same law, their centered coordinate variables have the same variances. The n -th coordinate of $B_{\mathbf{w}}X$ is $w_{n+1}X_{n+1}$, so

$$a_n = |w_{n+1}|^2 a_{n+1} \quad (n \geq 0).$$

If $a_0 = 0$, then the zeroth coordinate is μ -almost surely constant, so the support of μ is contained in a proper closed affine hyperplane of c_0 , contradicting full support. Hence $a_0 > 0$. In particular, no weight can be zero, and iteration gives

$$a_n = \frac{a_0}{|W_n|^2} \quad (n \geq 0).$$

Because $X \in c_0$ almost surely and $m \in c_0$, we have $Y_n \rightarrow 0$ almost surely, hence in probability. By the lemma, $a_n \rightarrow 0$. Therefore

$$\frac{a_0}{|W_n|^2} \rightarrow 0,$$

and since $a_0 > 0$, this is exactly $|W_n| \rightarrow \infty$.

For every $\lambda \in \mathbb{T}$, the vector

$$E(\lambda) = \left(\frac{\lambda^n}{W_n} \right)_{n \geq 0}$$

belongs to c_0 and is nonzero, and the recursion shows

$$B_{\mathbf{w}}E(\lambda) = \lambda E(\lambda).$$

Thus every $\lambda \in \mathbb{T}$ is a unimodular eigenvalue. The source question asks whether an ergodic Gaussian weighted shift on c_0 can avoid such eigenvalues; the stronger invariant full-support statement just proved rules this out. $\square$

Verification and Scope

The proof uses only invariance of a full-support Gaussian measure, not ergodicity. It also does not use strong mixing or weak mixing. The conclusion is therefore stronger than the negative answer required for Question (7).

The argument is specific to weighted backward shifts on c_0 . It does not answer the broader Question (6) asking on which Banach spaces one can find ergodic Gaussian operators without eigenvalues, nor does it settle Question (9) about Haar-nullity of the hypercyclic set for all ergodic weighted shifts.

Bounded novelty check: local run indexes were searched for arXiv:1112.1289 and for the phrases “ergodic weighted shift”, “ c_0 ”, and “unimodular eigenvalues”. Web searches on 2026-06-20 for exact variants of the source question did not reveal a later explicit answer.

References

- [1] F. Bayart and E. Matheron, *Mixing operators and small subsets of the circle*, arXiv:1112.1289.